Lecture 1

The particle in a box

Objectives. *A chain with fixed ends suggests the energy rule for a particle in a box. We build the self-adjoint Hamiltonian from the sine basis and prove that its energy postulate determines it uniquely. Energy readings remain defined for states outside its domain. Its evolution is Lecture 2's subject.*

- Compute the box energies and their reading probabilities.
- Prove self-adjointness by testing the adjoint on a basis.
- Distinguish finite mean energy from finite mean square energy.

1.1 From a chain to the box

A particle that moves along a line can be found anywhere in a continuum of positions. To find its energy, we start from a chain of sites, a system like those of *Quantum Mechanics I*, and let the sites come closer.

Take $N-1$ sites at the points $x_j = ja$ for $j = 1, \dots, N-1$, with the spacing $a = L/N$. The ends $x_0 = 0$ and $x_N = L$ are not sites. The state in which the particle sits on the site j is the unit vector e_j of $\mathbb{C}^{N-1}$. As on the ring of *Quantum Mechanics I*, Lecture 8, the particle hops between neighbouring sites with the hopping amplitude $\kappa > 0$. We give each site the on-site energy 2κ. This puts the bottom of the ring's band, $c - 2\kappa$ in *Quantum Mechanics I*, Lecture 8, at 0, and it keeps each level finite in the limit taken below. On $\psi = \sum_j \psi_j e_j$ in $\mathbb{C}^{N-1}$, the Hamiltonian H_N of this *open chain* acts by

$$(H_N\psi)_j = \kappa\,(2\psi_j - \psi_{j+1} - \psi_{j-1}) \quad (j = 1, \dots, N-1), \qquad \text{with } \psi_0 = \psi_N = 0, \tag{1.1}$$

since the particle cannot hop onto an end. In the site basis its matrix is

$$(H_N)_{jk} = \kappa(2\delta_{jk} - \delta_{j,k+1} - \delta_{j,k-1}) \quad (1 \le j, k \le N-1).$$

For example,

$$H_4 = \kappa \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

For $N = 2$, the matrix is (2κ).

Proposition 1.1 (The open chain). *For $n = 1, \dots, N-1$, the vector v_n with the components $(v_n)_j = \sin(n\pi j/N)$ is an eigenvector of H_N with the eigenvalue*

$$\varepsilon_n = 4\kappa \sin^2 \frac{n\pi}{2N}. \tag{1.2}$$

These $N-1$ eigenvalues are distinct, so they are all the levels of the chain.

Proof. Put $\theta = n\pi/N$. By the addition formula for the sine, $\sin\theta(j+1) + \sin\theta(j-1) = 2\cos\theta\,\sin\theta j$. So

$$\kappa\big(2\sin\theta j - \sin\theta(j+1) - \sin\theta(j-1)\big) = 2\kappa(1-\cos\theta)\sin\theta j = 4\kappa\sin^2\frac{\theta}{2}\,\sin\theta j.$$

At $j = 1$ and $j = N-1$, (1.1) uses $\psi_0 = 0$ and $\psi_N = 0$, and the formula $\sin\theta j$ gives these values at $j = 0$ and $j = N$, since $\sin 0 = 0$ and $\sin n\pi = 0$. So $H_N v_n = \varepsilon_n v_n$, and $v_n \neq 0$ because $(v_n)_1 = \sin(n\pi/N) > 0$. The angles $n\pi/2N$ lie in $(0, \pi/2)$, where $\sin^2$ increases, so the ε_n are distinct. A matrix of size $N-1$ has at most $N-1$ eigenvalues, so these are all of them. ∎

Now let the sites come closer in the fixed interval $[0, L]$. Give the particle the mass m, and take $\kappa = \hbar^2/2ma^2$. Let ψ be twice continuously differentiable on $[0, L]$, with the samples $\psi_j = \psi(x_j)$, and let $0 < x < L$. Then

$$\begin{aligned}\kappa\big(2\psi(x) - \psi(x+a) - \psi(x-a)\big) &= -\frac{\hbar^2}{2m}\,\frac{\psi(x+a) - 2\psi(x) + \psi(x-a)}{a^2} \\ &\longrightarrow -\frac{\hbar^2}{2m}\,\psi''(x) \quad \text{as } a \to 0,\end{aligned}$$

by Taylor's theorem applied to the real and imaginary parts of ψ. Indeed, let f be one of these parts, and let $0 < a < \min(x, L-x)$. Then there are points $y_+ \in (x, x+a)$ and $y_- \in (x-a, x)$ with

$$f(x+a) = f(x) + af'(x) + \frac{1}{2}a^2 f''(y_+), \qquad f(x-a) = f(x) - af'(x) + \frac{1}{2}a^2 f''(y_-).$$

Adding the two gives $(f(x+a) - 2f(x) + f(x-a))/a^2 = \frac{1}{2}\big(f''(y_+) + f''(y_-)\big)$, which tends to $f''(x)$ as $a \to 0$, since f'' is continuous. The hopping becomes a second derivative. The fixed ends $\psi_0 = \psi_N = 0$ sample a function only if $\psi(0) = \psi(L) = 0$. The levels converge too. At a fixed n, as $N \to \infty$, the limit $\sin t/t \to 1$ gives

$$\varepsilon_n = \frac{2\hbar^2 N^2}{mL^2}\,\sin^2\frac{n\pi}{2N} \longrightarrow \frac{\hbar^2\pi^2 n^2}{2mL^2},$$

from below, since $\sin t \leq t$. The components of the eigenvector v_n are the values of $\sin(n\pi x/L)$ at the sites.

Figure 1.1 compares the levels of the chain with this limit for $N = 4$, 8, 16 and 32.

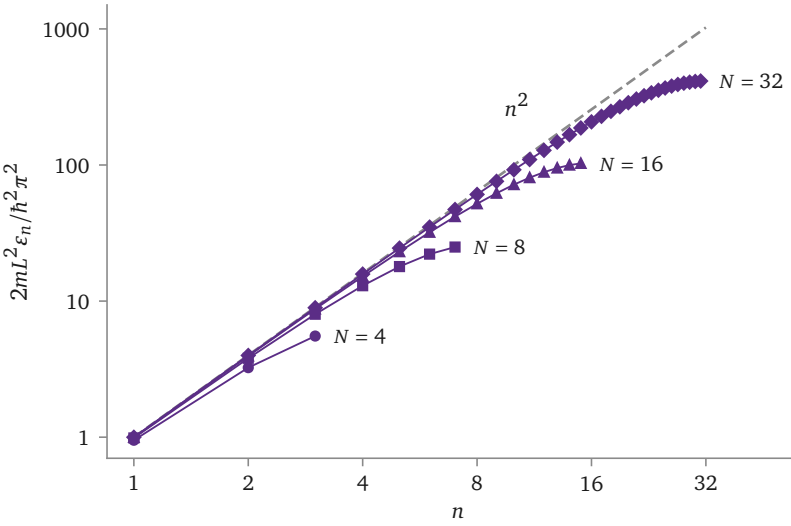


Figure 1.1: The levels ε_n of the open chain, $n = 1, \ldots, N-1$, for $N = 4$, 8, 16 and 32, in units of $\hbar^2\pi^2/2mL^2$, and the limit n^2 (dashed). Both axes are logarithmic. At a fixed n the levels approach the limit from below as N grows, and each chain's levels fall further below it as n approaches N.

The chain does not prove anything about a particle in a box. A limit of matrices of growing size is not yet an operator, and the walls were put in by hand, as the fixed ends. The chain motivates the energy postulate of § 1.5. The ring of *Quantum Mechanics I* has a different limit, a particle on a circle (Lecture 3).

Self-check 1.1

1. For $N = 3$, write the matrix of H_N and find its levels and eigenvectors. Compare each level with $\hbar^2\pi^2n^2/2mL^2$.

1.2 Wavefunctions in a box

The chain's state was a vector of amplitudes, one for each site. A state of the particle assigns an amplitude $\psi(x)$ to each position x in the box. The function ψ is the *wavefunction* of the state. Lecture 2 postulates that $|\psi(x)|^2$ is the probability density for finding the particle near x. The wavefunctions form a vector space with the inner product

$$\langle \varphi, \psi \rangle = \int_0^L \overline{\varphi(x)}\, \psi(x)\, \mathrm{d}x, \tag{1.3}$$

which is antilinear in the first slot, as in *Quantum Mechanics I*. The space is $L^2(0, L)$. Its elements are the functions with $\int_0^L |\psi|^2\, \mathrm{d}x < \infty$, and two functions that differ only on a set of measure zero count as the same element (*Linear Algebra*, Def. 13.15). With the norm $\|\psi\| = \langle \psi, \psi \rangle^{1/2}$, the space $L^2(0, L)$ is a Hilbert space, a complete inner product space.

So an element of $L^2(0, L)$ has no value at a single point. Changing ψ at $x = 0$ alone changes no integral, and it does not change the element. A condition such as $\psi(0) = 0$ therefore cannot be imposed on the states. It will return as a condition on the domain of the Hamiltonian.

Postulate 1.2 (States of a particle in a box)**.** The states of a particle in the box are the rays of $L^2(0, L)$.

Every state is represented by a sequence of amplitudes, the coefficients of a unit vector along the functions

$$u_n(x) = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \qquad (n = 1, 2, 3, \dots). \tag{1.4}$$

The functions u_n are orthonormal. Indeed, $2 \sin\alpha \sin\beta = \cos(\alpha - \beta) - \cos(\alpha + \beta)$, and $\int_0^L \cos(j\pi x/L)\,\mathrm{d}x$ is L for $j = 0$ and 0 for every other integer j. With $\alpha = n\pi x/L$ and $\beta = k\pi x/L$, where $n, k \geq 1$, this gives $\langle u_n, u_k\rangle = \int_0^L u_n u_k\,\mathrm{d}x = \delta_{nk}$.

Theorem 1.3 (The sine basis · *cited*)**.** *The functions u_n form a complete orthonormal basis of $L^2(0, L)$. Every $\psi \in L^2(0, L)$ is the norm-convergent series*

$$\psi = \sum_{n=1}^{\infty} c_n u_n, \qquad c_n = \langle u_n, \psi\rangle\,, \tag{1.5}$$

and every sequence (c_n) with $\sum_n |c_n|^2 < \infty$ is the coefficient sequence of exactly one ψ. If $a_n = \langle u_n, \varphi\rangle$, then

$$\langle \varphi, \psi\rangle = \sum_{n=1}^{\infty} \overline{a_n}\, c_n \quad \textit{and} \quad \|\psi\|^2 = \sum_{n=1}^{\infty} |c_n|^2\,. \tag{1.6}$$

This is the Fourier sine series (Hall, Prop. 3.24). The general facts about complete bases are *Linear Algebra*, Thms 13.23 and 13.24, and the second identity in (1.6) is *Parseval's identity*. The sine basis theorem is the standing fact S2 of § 0.3, taken from *Linear Algebra* without proof, so a result that uses it counts as proved. So a state of the box is represented by a sequence of infinitely many amplitudes $c_1, c_2, \dots$ with $\sum_n |c_n|^2 = 1$, up to a common phase, and every such sequence represents a state.

Self-check 1.2

1. Expand $\psi(x) = \sin^3(\pi x/L)$ in the u_n, and find $\|\psi\|^2$.

1.3 Stationary waves

The chain of § 1.1 suggests *Schrödinger's rule*: for a free particle of mass m, the energy acts on a wavefunction by $\psi \mapsto -(\hbar^2/2m)\,\psi''$. We take the rule as given here, and § 1.5 makes it a postulate. A wavefunction of definite energy E, a *stationary wave*, solves $-(\hbar^2/2m)\,u'' = Eu$. Impenetrable walls are modelled by asking the stationary wave to vanish at both walls, as the chain's fixed ends suggest. Hall (§3.9) motivates this condition by a very deep well.

Lemma 1.4 (Stationary waves). *Let E be a real number. A nonzero, twice continuously differentiable u on $[0, L]$ with*

$$-\frac{\hbar^2}{2m}u'' = Eu \qquad \text{and} \qquad u(0) = u(L) = 0 \tag{1.7}$$

exists if and only if E is one of the energies

$$E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2} \qquad (n = 1, 2, 3, \dots). \tag{1.8}$$

For $E = E_n$, the solutions are the multiples of u_n.

Proof. First let $E \le 0$. Multiplying (1.7) by $\overline{u}$ and integrating by parts, with $u = 0$ at both walls, gives

$$E\int_0^L |u|^2 \, dx = \frac{\hbar^2}{2m}\int_0^L |u'|^2 \, dx.$$

The left side is nonpositive and the right side is nonnegative, so $u' = 0$. The boundary value then gives $u = 0$.

For $E > 0$, put $k = \sqrt{2mE}/\hbar$. The solutions are $u = a\sin kx + b\cos kx$. The wall at 0 forces $b = 0$, and the wall at L forces $\sin kL = 0$, that is, $kL = n\pi$ for an integer $n \ge 1$, and $E = \hbar^2k^2/2m = E_n$. Conversely, each u_n satisfies both the differential equation and the boundary conditions. This gives all the solutions. ∎

The stationary waves are the sine functions u_n of the sine basis (Figure 1.2), and their energies E_n are the limits of the chain's levels. A classical particle in the box may have any energy $E \ge 0$. The quantum particle has only the energies E_n, and even the least of them, $E_1 = \hbar^2\pi^2/2mL^2$, is positive.

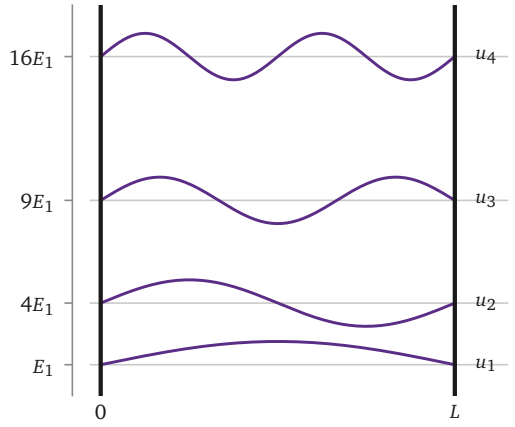


Figure 1.2: The stationary waves $u_1, \dots, u_4$ between the walls, each drawn about the height of its energy $E_n = n^2E_1$. The vertical size of each wave is not to scale.

Self-check 1.3

1. Find E_1 in electronvolts for an electron in a box of length $L = 1$ nm.

The physics

In a cyanine dye, a chain of carbon atoms carries electrons, the π electrons, that move along the whole chain. Its bonds are drawn as alternating single and double bonds, but they are of nearly equal length. Kuhn modelled these electrons in 1949 as free particles in a box of the chain's length L. With $2K$ such electrons filling the lowest K levels, two to a level (*Quantum Mechanics I*, Lecture 14), the least energy that the electrons can absorb lifts one of them from the level K to the level $K+1$, and it is $E_{K+1} - E_K = (2K+1)E_1$. For $K = 5$ and $L = 1.4\,\text{nm}$ the model predicts the absorption of light of wavelength about 590 nm, which is yellow. Such dyes do absorb visible light, the longer chains at longer wavelengths. The model works less well for chains whose bond lengths alternate, such as those of the polyenes (Autschbach, 2007).

1.4 Why the energy needs a domain

The energies grow without bound, $E_n = n^2 E_1$, and this is where the new difficulty comes from. In the coefficients of (1.5), an energy that multiplies each stationary wave u_n by E_n multiplies each amplitude c_n by E_n. For some states the result is not square-summable, so it is the coefficient sequence of no vector of $L^2(0, L)$. § 1.7 gives such a state, with $c_n \propto 1/n^2$. The energy multiplies the unit vector u_n by E_n, so it is not bounded. By the Hellinger–Toeplitz theorem of § 0.1, it cannot be a symmetric operator defined on every vector.

Recall from Lecture 0 the operators with a domain (Definition 0.1), the adjoint $A^\dagger$ of a densely defined operator A (Definition 0.2), and the symmetric and the self-adjoint operators (Definition 0.3). A symmetric operator A has $D(A) \subseteq D(A^\dagger)$, with $A^\dagger\varphi = A\varphi$ there, and it is self-adjoint when the two domains are equal.

Postulate 1.5 (Observables with an eigenbasis)**.** The observables of a particle in the box are self-adjoint operators on $L^2(0, L)$. Suppose an observable A has an orthonormal basis of eigenvectors $e_1, e_2, \ldots$, with $Ae_k = a_k e_k$. Then its readings are the numbers a_k, and the measurement postulate of *Quantum Mechanics I* (Born's and Lüders' rules, Postulate 4.1) holds, with the projection E_a (not an energy)

$$E_a\psi = \sum_{k:\, a_k = a} \langle e_k, \psi\rangle\, e_k, \qquad \text{Prob}(a) = \|E_a\psi\|^2 = \sum_{k:\, a_k = a} |\langle e_k, \psi\rangle|^2 . \tag{1.9}$$

By Parseval's identity, the probabilities add to 1. This is where completeness matters. If the basis missed a direction, part of the probability would be lost.

The postulate asks for self-adjoint operators, not only symmetric ones, for two reasons that later lectures give. By the spectral theorem, which Lecture 5 cites, exactly the self-adjoint operators are integrals of spectral measures, the rules that give the probabilities of readings in intervals. The evolutions in time are strongly continuous unitary groups. Lecture 7 proves that every such group is generated by a self-adjoint operator divided by $\hbar$, and only sketches the converse, that every self-adjoint operator divided by $\hbar$ generates such a group.

1.5 The box Hamiltonian

Postulate 1.6 (The energy in a box)**.** The energy of a particle of mass m in the box is an observable H. On every twice continuously differentiable ψ on $[0, L]$ with $\psi(0) = \psi(L) = 0$, it acts by Schrödinger's rule

$$H\psi = -\frac{\hbar^2}{2m}\psi''. \tag{1.10}$$

The postulate fixes H on a small set of functions. Does it fix H? We first build one self-adjoint operator that obeys it, from the coefficients.

Definition 1.7 (The box Hamiltonian)**.** With $c_n = \langle u_n, \psi\rangle$, the *box Hamiltonian* H has the domain and the action

$$D(H) = \left\{\psi \in L^2(0, L) : \sum_{n=1}^{\infty} n^4 |c_n|^2 < \infty\right\}, \qquad H\psi = \sum_{n=1}^{\infty} E_n c_n u_n. \tag{1.11}$$

Since $E_n = n^2 E_1$, the condition on the domain says $\sum_n E_n^2 |c_n|^2 < \infty$. By the sine basis theorem, this is exactly the condition for the series $\sum_n E_n c_n u_n$ to converge. In the coefficients, H is the diagonal operator $(c_n) \mapsto (E_n c_n)$.

Theorem 1.8 (The box Hamiltonian)**.** *The box Hamiltonian H is densely defined and self-adjoint. Its eigenvalues are exactly the energies E_n, and the eigenvectors for E_n are the nonzero multiples of u_n. It obeys the energy postulate: every twice continuously differentiable ψ on $[0, L]$ with $\psi(0) = \psi(L) = 0$ lies in $D(H)$, and $H\psi = -(\hbar^2/2m)\psi''$. It is the only self-adjoint operator that obeys the energy postulate.*

As for the diagonal operator of Example 0.6, testing the adjoint's condition on the vectors u_n fixes every coefficient of χ.

Proof.

1. *Dense.* A finite combination of the u_n lies in $D(H)$, since its sum $\sum_n n^4 |c_n|^2$ has finitely many nonzero terms. By the sine basis theorem, the partial sums $\sum_{n \le M} c_n u_n$ of any ψ converge to ψ. So $D(H)$ is dense.
2. *Symmetric.* Let $\varphi, \psi \in D(H)$, with coefficients a_n and c_n. Since each E_n is real, (1.6) gives
$$\langle \varphi, H\psi\rangle = \sum_n \overline{a_n}\, E_n c_n = \sum_n \overline{E_n a_n}\, c_n = \langle H\varphi, \psi\rangle .$$
3. *No larger adjoint.* Let $\varphi \in D(H^\dagger)$, with $H^\dagger\varphi = \chi$, and put $a_n = \langle u_n, \varphi\rangle$. Take $\psi = u_n$ in (0.1). The left side is $\langle \varphi, Hu_n\rangle = E_n \langle \varphi, u_n\rangle = E_n\overline{a_n}$, and the right side is $\langle \chi, u_n\rangle = \overline{\langle u_n, \chi\rangle}$. So $\langle u_n, \chi\rangle = E_n a_n$ for every n. By Parseval's identity, $\sum_n E_n^2 |a_n|^2 = \|\chi\|^2 < \infty$. Hence $\varphi \in D(H)$ and $\chi = \sum_n E_n a_n u_n = H\varphi$. So $D(H^\dagger) \subseteq D(H)$. By the symmetric step and the inclusion for symmetric operators recalled in § 1.4, the opposite inclusion holds, with $H^\dagger\varphi = H\varphi$. So $H^\dagger = H$.
4. *Eigenvalues.* By definition, $Hu_n = E_n u_n$. Conversely, let $H\psi = \lambda\psi$ with $\psi \neq 0$. Comparing coefficients gives $E_n c_n = \lambda c_n$ for every n. Some c_n is nonzero, so $\lambda = E_n$. The energies are distinct, so $c_j = 0$ for every $j \neq n$, and $\psi = c_n u_n$.
5. *The rule.* Let ψ be twice continuously differentiable on $[0, L]$, with $\psi(0) = \psi(L) = 0$.

Since the u_n are real, integrating by parts twice gives

$$\langle u_n, -\psi'' \rangle = \langle -u_n'', \psi \rangle + \Big[u_n' \psi - u_n \psi' \Big]_0^L .$$

The bracket vanishes, because u_n and ψ both vanish at 0 and at L. Since $-u_n'' = (n\pi/L)^2 u_n$, the coefficients of $-(\hbar^2/2m)\,\psi''$ are $E_n c_n$. The function ψ'' is continuous on $[0, L]$, so it lies in $L^2(0, L)$, and Parseval's identity gives $\sum_n E_n^2 \, |c_n|^2 = (\hbar^2/2m)^2 \, \|\psi''\|^2 < \infty$. Hence $\psi \in D(H)$, and $H\psi = \sum_n E_n c_n u_n = -(\hbar^2/2m)\,\psi''$.

6. *The only one.* Let B be a self-adjoint operator that obeys the energy postulate. Each u_n is twice continuously differentiable and vanishes at the walls, so $Bu_n = -(\hbar^2/2m)\,u_n'' = E_n u_n$. For $\varphi \in D(B)$, the symmetry of B gives $\langle u_n, B\varphi \rangle = \langle Bu_n, \varphi \rangle = E_n a_n$, and the step on the adjoint, with $B\varphi$ in place of χ, gives $\varphi \in D(H)$ and $B\varphi = H\varphi$. So $B \subseteq H$, and since adjoints reverse inclusions (Proposition 0.5), $H = H^\dagger \subseteq B^\dagger = B$. Hence $B = H$. ∎

So the energy postulate determines the Hamiltonian. Its domain holds more than the twice continuously differentiable functions that vanish at the walls (§ 1.7), and Lecture 3 describes it by derivatives. The condition $\psi(0) = \psi(L) = 0$ is the *Dirichlet condition*. The walls are a condition on the domain of H, not on the states.

This condition uses values at points, and an element of $L^2(0, L)$ has none. It makes sense on $D(H)$, because every vector of $D(H)$ has a continuous representative, that is, a continuous function among the functions that count as this element. Let $\psi \in D(H)$. By the Cauchy–Schwarz inequality,

$$\sum_n |c_n| \le \Big(\sum_n n^{-4} \Big)^{1/2} \Big(\sum_n n^4 \, |c_n|^2 \Big)^{1/2} < \infty .$$

Since $|u_n(x)| \le \sqrt{2/L}$ for every x, the series $\sum_n c_n u_n$ converges uniformly on $[0, L]$. Its sum is a continuous function, and it vanishes at both walls, since every u_n does. Uniform convergence on a bounded interval implies convergence in norm, so this function represents ψ. Two continuous functions that differ only on a set of measure zero are equal, so this representative is unique, and the condition is read on it. Lecture 3 describes these functions by their derivatives.

Self-check 1.5

1. A self-adjoint operator B has $Bu_n = E_n u_n$ for every n. Must B be H?

1.6 Energy readings in a parabola

Example 1.9 (A parabola in the box).

1. *Prepare.* The state is

$$\psi(x) = \sqrt{\frac{30}{L^5}}\, x(L - x).$$

It vanishes at both walls. Since $\int_0^L x^2 (L - x)^2 \, \mathrm{d}x = L^5/30$, it is a unit vector.

2. *Find the coefficients.* The function ψ is twice continuously differentiable and vanishes at the walls. So the step on the rule in the proof of Theorem 1.8 gives $(n\pi/L)^2 c_n = \langle u_n, -\psi'' \rangle$. Here $-\psi'' = 2\sqrt{30/L^5}$ is a constant, and

$$\langle u_n, 1 \rangle = \sqrt{\frac{2}{L}} \int_0^L \sin \frac{n\pi x}{L} \, \mathrm{d}x = \sqrt{\frac{2}{L}}\, \frac{L\,(1 - (-1)^n)}{n\pi}.$$

Hence

$$c_n = \frac{8\sqrt{15}}{\pi^3 n^3} \quad \text{for odd } n, \qquad c_n = 0 \quad \text{for even } n.$$

3. *Read the energies.* By the observables postulate, a measurement of the energy gives E_n with probability c_n^2, and it leaves the state u_n. So $\text{Prob}(E_1) = 960/\pi^6 \approx 0.9986$ and $\text{Prob}(E_3) = 960/(729\pi^6) \approx 0.0014$. The even levels never occur, because the state is symmetric about the midpoint $L/2$ and the u_n with even n are antisymmetric about it. By Parseval's identity the probabilities add to 1.
4. *Apply H.* By Theorem 1.8, $\psi \in D(H)$, and

$$H\psi = -\frac{\hbar^2}{2m}\psi'' = \frac{\hbar^2}{m}\sqrt{\frac{30}{L^5}},$$

a constant function. It does not vanish at the walls. The walls are a condition on ψ, not on $H\psi$.

5. *Find the mean energy.* By the inner product formula (1.6) and the previous step, the mean of the energy readings is

$$\sum_n E_n \,\text{Prob}(E_n) = \langle \psi, H\psi \rangle = \frac{\hbar^2}{m}\sqrt{\frac{30}{L^5}} \int_0^L \psi(x)\,dx = \frac{5\hbar^2}{mL^2} = \frac{10}{\pi^2} E_1 \approx 1.013\, E_1.$$

The mean lies just above the least energy E_1, as the probabilities in the third step predict.

Self-check 1.6

1. Compute $\int_0^L u_1(x)\, x\, u_2(x)\,dx$ and $\int_0^L u_1(x)\, x\, u_3(x)\,dx$. Why does the second vanish?
2. Take the state with $c_1 = c_3 = 1/\sqrt{2}$ and all other $c_n = 0$. Find $\text{Prob}(E_3)$ and the mean energy.

1.7 A state outside the domain

Some states lie outside the domain of H. Take the coefficients

$$c_n = \frac{\sqrt{90}}{\pi^2 n^2} \qquad (n = 1, 2, 3, \dots).$$

By Euler's sum $\sum_n 1/n^4 = \pi^4/90$, the coefficients define a unit vector $\psi = \sum_n c_n u_n$. But $n^4 c_n^2 = 90/\pi^4$ for every n, so $\sum_n n^4 c_n^2$ diverges, and $\psi \notin D(H)$. The vector $H\psi$ is not defined.

Figure 1.3 draws this state beside the parabola of Example 1.9. Both curves are humps that vanish at both walls, and a look does not settle which of them lies in $D(H)$. The coefficients do. For the parabola, $n^4 c_n^2 = 960/\pi^6 n^2$ on the odd n, so the sum converges. For this state, every term of the sum is $90/\pi^4$. Lecture 3 shows that every vector of $D(H)$ has a continuous derivative on $[0, L]$, which the infinite slope of this state at the left wall rules out.

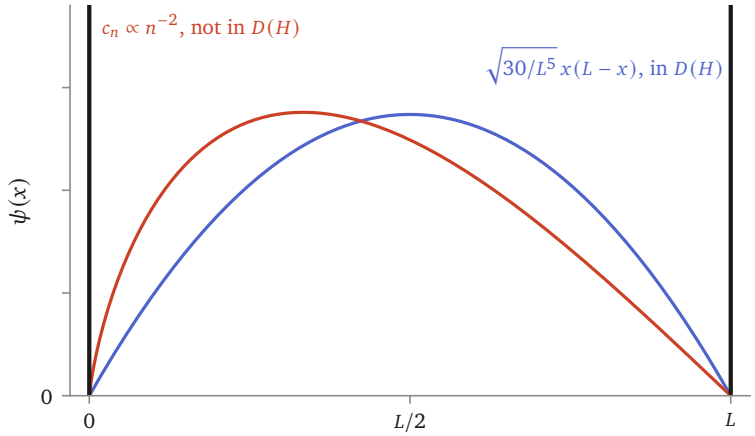


Figure 1.3: Two unit vectors between the walls. The parabola of Example 1.9, with $c_n = 8\sqrt{15}/\pi^3 n^3$ on the odd n, lies in $D(H)$. The state with $c_n = \sqrt{90}/\pi^2 n^2$ for every n does not. Its curve has an infinite slope at the left wall and a finite slope at the right wall. Membership in the domain is the condition $\sum_n n^4 |c_n|^2 < \infty$ on the coefficients.

The energy readings of ψ are nevertheless defined, by the observables postulate. By Euler's sum $\sum_n 1/n^2 = \pi^2/6$, the mean energy is

$$\sum_n E_n \operatorname{Prob}(E_n) = \frac{90}{\pi^4} E_1 \sum_n \frac{1}{n^2} = \frac{15}{\pi^2} E_1 \approx 1.52\, E_1,$$

which is finite. The domain is therefore not a set of allowed states. Every unit vector of $L^2(0, L)$ represents a state, its energy can be measured, and Lecture 2 gives its time evolution.

The domain records a moment of the energy readings. For $\psi \in D(H)$, Parseval's identity gives

$$\|H\psi\|^2 = \sum_n E_n^2 |c_n|^2 = \sum_n E_n^2 \operatorname{Prob}(E_n),$$

the mean square of the energy readings. So a state lies in $D(H)$ exactly when its energy readings have a finite mean square. For the state above the mean is finite and the mean square is not.

The rule alone is not self-adjoint. Symmetry alone does not make an observable. Physics texts often call an operator *Hermitian* when it is symmetric. In infinite dimension, symmetric and self-adjoint are different conditions. Integration by parts, as in the step on the rule, proves that Schrödinger's rule on the twice continuously differentiable functions that vanish at the walls is symmetric. It is not self-adjoint there.

We show this by finding a vector of $D(H)$ outside the domain of the rule. Let g be the function equal to 1 on $[0, L/2]$ and to 0 on $(L/2, L)$, and let ψ have the coefficients $c_n = \langle u_n, g\rangle / E_n$. By Parseval's identity,

$$\sum_n E_n^2 |c_n|^2 = \|g\|^2 < \infty,$$

so ψ lies in $D(H)$, and $H\psi = g$. This vector is the function

$$\psi(x) = \frac{2m}{\hbar^2} \begin{cases} 3Lx/8 - x^2/2 & \text{for } 0 \le x \le L/2, \\ L(L-x)/8 & \text{for } L/2 \le x \le L. \end{cases}$$

To see this, note that this function vanishes at the walls, has a continuous derivative, and satisfies $-(\hbar^2/2m)\psi'' = g$ on each half of the box. Two integrations by parts on each half, as in the step on the rule, give its coefficients $\langle u_n, g\rangle / E_n = c_n$. The terms at $L/2$ cancel, since the function and its derivative are continuous there. So, by Theorem 1.3, this function is ψ.

The second derivative of this function jumps at $L/2$. If ψ had a twice continuously differentiable representative that vanishes at the walls, the step on the rule would give $H\psi = -(\hbar^2/2m)\,\psi''$, and $H\psi$ would have a continuous representative. The function g jumps at $L/2$, so it has none. Hence the rule is a proper restriction of H: by Theorem 1.8 the operator H extends it, and $D(H)$ is larger. If the rule were self-adjoint, it would obey the energy postulate, and by the uniqueness in Theorem 1.8 it would equal H. So it is not self-adjoint.

Lecture 3 shows another symmetric operator in this box that is not self-adjoint, the momentum with walls.

Self-check 1.7

1. The coefficients $c_n = \sqrt{6}/(\pi n)$ define a unit vector. Is it in $D(H)$? Is its mean energy finite?

The particle in a box, as a quantum system

States: the rays of $L^2(0, L)$, represented by the sequences of coefficients $c_n = \langle u_n, \psi\rangle$ with $\sum_n |c_n|^2 = 1$, up to a common phase (Postulate 1.2 and Theorem 1.3). *Observables:* self-adjoint operators, each with its domain (Postulate 1.5 and Definition 0.3). *Energy:* H multiplies c_n by $E_n = \hbar^2\pi^2n^2/2mL^2$ on the vectors with $\sum_n n^4\,|c_n|^2 < \infty$. It is self-adjoint, and it is the only self-adjoint operator that obeys the energy postulate (Postulate 1.6 and Theorem 1.8). *Measurement:* the energy reads E_n with probability $|c_n|^2$, and the state afterwards is u_n (Postulate 1.5).

Summary.

- *The open chain*: with $\kappa = \hbar^2/2ma^2$, its levels $4\kappa \sin^2(n\pi/2N)$ approach the box's energies $E_n = n^2E_1$ at each fixed n as the sites come closer (Proposition 1.1).
- *The box Hamiltonian* is self-adjoint on its domain, and its eigenvalues are the E_n. It is the only self-adjoint operator that obeys the energy postulate (Theorem 1.8).
- *A state outside the domain*: the domain holds the states whose energy readings have a finite mean square.
- *The rule alone is not self-adjoint*: on the twice continuously differentiable functions that vanish at the walls, it is symmetric, and it is a proper restriction of H (§ 1.7).

Lecture 2 lets the box Hamiltonian evolve the states, and asks where the particle is.

Exercises. Exercise Sheet 1.

Self-check answers.

1.1 1. The rows are $(2\kappa, -\kappa)$ and $(-\kappa, 2\kappa)$. The levels are κ, with $(1, 1)$, and 3κ, with $(1, -1)$. With $\kappa = 9\hbar^2/2mL^2$ the ratios are $9/\pi^2 \approx 0.912$ for $n = 1$ and $27/4\pi^2 \approx 0.684$ for $n = 2$.

1.2 1. By $\sin^3 y = \frac{1}{4}(3\sin y - \sin 3y)$, $\psi = \frac{1}{4}\sqrt{L/2}\,(3u_1 - u_3)$, and $\|\psi\|^2 = 5L/16$.

1.3 1. $E_1 \approx 0.376\,\text{eV}$.

1.5 1. Yes. The step "the only one" of the proof of Theorem 1.8 used only that B is self-adjoint and that $Bu_n = E_nu_n$ for every n.

1.6 1. $-16L/9\pi^2$ and 0. Write $x = (x - L/2) + L/2$. The term with $L/2$ gives $(L/2)\,\langle u_1, u_3\rangle = 0$. The functions u_1 and u_3 are symmetric about $L/2$ and $x - L/2$ is antisymmetric, so the other term vanishes too. 2. $\text{Prob}(E_3) = \frac{1}{2}$. The mean energy is $\frac{1}{2}E_1 + \frac{1}{2} \cdot 9E_1 = 5E_1$.

1.7 1. No: $n^4 c_n^2 = 6n^2/\pi^2$. The mean energy $\sum_n E_1 n^2 \cdot 6/\pi^2 n^2$ is infinite.

References

Hall, *Quantum Theory for Mathematicians*, §3.9 (pp. 80–82). *Linear Algebra for Physics*, Def. 13.15, Thms 13.23 and 13.24. *Quantum Mechanics I*, Postulate 4.1, Lectures 8 and 14. Kuhn, *Journal of Chemical Physics* 17, 1198 (1949). Autschbach, *Journal of Chemical Education* 84, 1840 (2007).